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Innovation and Production in the Global Economy

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1 Big picture

- **Question.** How does globalization change the global allocation of ideas and manufacturing? More precisely, when multinational production (MP) lets firms separate where ideas are created from where goods are produced, which countries specialize in innovation, which countries specialize in production, and what are the welfare and wage consequences?
- **Core mechanism.** Firms innovate in a home country, draw productivity levels for possible production locations, and then choose where to produce for each destination market. Trade costs encourage production near consumers, while MP costs discourage using foreign production locations. Fixed innovation costs and market size generate home-market effects in innovation.
- **Main empirical fact.** Countries with high manufacturing R&D intensity tend to have high net outward multinational production: they are homes of firms whose foreign affiliate sales exceed sales of foreign affiliates located in the country. The United States also simultaneously increased manufacturing R&D intensity and the foreign-affiliate employment share of US multinationals.
- **Model contribution.** The paper builds a quantifiable, multi-country, monopolistic-competition model with endogenous firm creation, bilateral trade costs, bilateral MP costs, and correlated productivity draws across production locations.
- **Calibration strategy.** The model is disciplined by aggregate trade flows, multinational production flows, manufacturing production, innovation shares, and a novel distinction between restricted and unrestricted gravity elasticities.
- **Key gravity prediction.** Trade flows restricted to parents and affiliates of firms from a given country should be more elastic to trade costs than unrestricted aggregate trade flows. The paper confirms this prediction using US BEA multinational-firm data.
- **Main welfare result.** All countries gain from overall openness, but the direct gains can be partly offset by indirect effects when specialization shifts innovation across countries. Some countries can lose from a marginal liberalization of trade or MP, even though losses are generally small.
- **Distributional result.** The model distinguishes innovation workers and production workers. Contrary to fears about offshoring, production workers in innovation-specializing countries can gain from MP liberalization because lower goods prices and equilibrium wage adjustments dominate the production relocation effect.
- **China result.** The integration of China affects innovation and production shares across countries. China can specialize more in production; countries with strong links to China may benefit, while some innovation activity can be reallocated across countries.
- **Takeaway.** Openness through trade and MP is not just about lower prices. It reorganizes where ideas are created, where goods are produced, and how income is divided between innovation and production tasks.

2 Incremental contribution of the paper

- **Endogenizes innovation and production jointly.** Earlier quantitative MP models usually treat firm origins, entry, or innovation as fixed. This paper makes innovation endogenous and allows production to be located separately from the country where the idea is created.
- **Connects multinational production to home-market effects.** The model shows that MP can magnify home-market effects in innovation. Large or well-positioned countries can become innovation hubs, while other countries specialize in production.
- **Introduces a tractable multivariate productivity structure.** The paper uses a multivariate Pareto distribution that lets productivity draws across production locations be correlated while preserving gravity-like aggregation.

- **Provides a new empirical target: restricted gravity.** By comparing trade flows within the network of US parents and affiliates to unrestricted trade flows, the paper obtains information about the correlation of productivity across production locations.
- **Moves beyond aggregate gains from openness.** The paper decomposes welfare into direct gains and indirect innovation-reallocation effects, and also separates gains from trade from gains from MP.
- **Links offshoring to worker types.** By distinguishing production workers from innovation workers, it evaluates whether workers in the production sector necessarily lose when firms move production abroad. The answer is no in the calibrated exercises.
- **Quantifies China integration in a global model.** The paper studies China not as an isolated import shock but as a change in global trade and MP opportunities that reallocates innovation and production across many countries.
- **Clarifies why trade and MP can be substitutes.** Once a country can access foreign varieties through trade, the additional gains from MP can be smaller, and vice versa. This helps explain why gains from trade or MP alone can be smaller than gains from total openness.

3 Data and measurement

3.1 Countries and aggregate variables

The quantitative model is calibrated to 26 countries with data on manufacturing trade, production, multinational production, and innovation-related activity. The objects are country-level absorption, output, bilateral trade shares, bilateral multinational production shares, production-worker income, and innovation-worker income.

Interpretation: The model is not estimated on a single bilateral flow or on a single country. It is designed to match a global matrix of trade and MP relationships, so a change in one country's MP cost affects many countries through general equilibrium.

3.2 Trade and multinational production

- Trade data come from manufacturing flows, including domestic absorption and bilateral import/export flows.
- MP data measure the gross value of production or sales of multinational affiliates by source and host country.
- Outward MP is production by a country's firms abroad. Inward MP is production in a country by foreign-owned affiliates.
- Net MP is outward affiliate sales minus inward affiliate sales, divided by their sum.

Interpretation: A country with high outward relative to inward MP is an idea-exporting or innovation-specializing country in the broad sense: its firms use foreign production locations to serve markets.

3.3 Innovation measurement

Innovation is measured using manufacturing R&D and R&D employment shares. The paper emphasizes that R&D is an imperfect but informative proxy for the broader activity of creating ideas, firm-specific intangible capital, marketing, and management technology.

Interpretation: The model’s innovation share is much larger than the observed R&D employment share because the model’s innovation sector includes all fixed-cost idea creation and intangible activities, not only measured laboratory R&D.

3.4 Trade and MP cost measurement

The model requires bilateral trade costs τ_{ln} and bilateral MP costs γ_{il} . These are calibrated from observed trade and MP shares using a generalized Head-Ries style procedure, together with restrictions implied by the model.

Interpretation: Trade costs govern where goods are shipped. MP costs govern where ideas can be used for production. The model’s central object is the combination of these two frictions.

3.5 Restricted versus unrestricted gravity

A key data moment is the difference between restricted and unrestricted gravity elasticities. Restricted gravity uses trade flows among the production locations of firms from the same source country. Unrestricted gravity uses aggregate bilateral trade flows.

Interpretation: If productivity across locations is correlated for a firm’s production possibilities, then restricted flows should respond more strongly to trade costs. This is why Table 1 is central to calibrating the correlation parameter.

4 Model and theoretical structure

4.1 Preferences and price index

Consumers have CES preferences over a continuum of differentiated goods, with elasticity of substitution $\sigma > 1$. The country i price index is

$$P_i = \left(\int_{\omega \in \Omega} p_i(\omega)^{1-\sigma} d\omega \right)^{\frac{1}{1-\sigma}}.$$

Interpretation: CES preferences imply a constant markup over delivered marginal cost and make aggregate welfare depend on the availability and delivered prices of differentiated varieties.

4.2 Innovation, production, and consumption locations

Each firm has an origin country i , chooses a production location l , and sells in a destination country n . The three-country notation is essential:

- i : country where the idea is created;
- l : country where the good is produced;
- n : market where the good is consumed.

For a firm from i producing in l and selling to n , costs include MP cost γ_{il} and trade cost τ_{ln} . The delivered price is proportional to

$$\gamma_{il} \tau_{ln} \frac{w_l^p}{z_l},$$

where w_l^p is the production wage and z_l is productivity in production location l .

Interpretation: The firm’s problem separates innovation from manufacturing. A US firm can innovate in the United States, produce in China, and sell in Europe if the productivity and trade/MP cost tradeoff makes that route cheapest.

4.3 Productivity draws and the multivariate Pareto

The productivity vector across production locations is drawn from a multivariate Pareto distribution. The parameter θ controls tail thickness, and ρ controls correlation across productivity draws. Low ρ means productivity advantages are location-specific; high ρ means a firm that is productive in one location tends to be productive elsewhere.

Interpretation: The correlation parameter matters because it controls whether a firm’s comparative advantage is attached to a particular production location or is more global. This affects the gap between restricted and unrestricted gravity elasticities.

4.4 Trade and MP shares

The model implies gravity-like expressions for bilateral trade and MP shares. Trade shares depend on trade costs and the set of firms serving market n from production country l . MP shares depend on MP costs and the ability of firms from country i to produce in country l .

Interpretation: The model is designed to be “quantifiable”: once trade shares, MP shares, and production data are observed, many trade and MP cost parameters can be recovered and then used for counterfactuals.

4.5 Innovation share and specialization

The innovation share r_i is the share of country i ’s labor income associated with innovation workers rather than production workers. Countries with $r_i > \eta$ are innovation-specializing; countries with $r_i < \eta$ are production-specializing, where η is the profit share.

Interpretation: Specialization is not binary. The innovation share summarizes how much of a country’s labor income is tied to idea creation instead of production activity.

4.6 Direct and indirect welfare effects

The paper decomposes gains from openness into direct and indirect components. The direct effect is the usual lower-cost access to goods and locations. The indirect effect captures how openness reallocates innovation across countries and therefore changes wages and terms of trade.

Interpretation: The indirect effect is the reason some countries can lose from a marginal liberalization of trade or MP even when total openness raises welfare for all countries.

5 Calibration, validation, and counterfactual design

5.1 Calibration targets

The calibration targets include the elasticity of substitution, the Fréchet shape parameter, restricted and unrestricted gravity elasticities, profit share, average productivities in innovation and production, bilateral

trade shares, and bilateral MP shares.

Interpretation: The calibration is disciplined by both aggregate cross-country data and firm-network trade data. This makes the counterfactuals informative about innovation and production specialization rather than merely matching aggregate trade volumes.

5.2 Model fit

The paper validates the calibration by comparing model-implied trade and MP shares to observed data. It evaluates both aggregate country-level flows and bilateral flows.

Interpretation: The model fits bilateral MP flows especially well, which matters because MP is the distinctive mechanism relative to standard trade models.

5.3 Counterfactuals

The main counterfactuals are:

- move from autarky to openness;
- isolate gains from trade and gains from MP;
- reduce all MP barriers by 5 percent;
- study China’s integration through China autarky, unilateral MP liberalization into China, and frictionless US MP into China;
- robustness exercises including alternative calibration and fixed-cost discussions.

Interpretation: The counterfactuals are not only about aggregate welfare. They show how openness changes specialization and the distribution of gains between production and innovation workers.

6 Identification and empirical design logic

6.1 What identifies the model’s key mechanisms

The paper is a structural quantitative model rather than a quasi-experimental design. Identification comes from matching multiple moments: trade shares, MP shares, restricted versus unrestricted gravity elasticities, and innovation shares.

6.2 Role of restricted gravity

Restricted gravity identifies how the productivity distribution differs from a model with independent production-location draws. If restricted trade is much more elastic to trade costs than unrestricted trade, then the model needs correlated productivity across locations.

6.3 Role of innovation shares

Innovation shares help discipline the location of idea creation. They allow the calibration to distinguish countries that specialize in innovation because of comparative advantage from countries that do so because of home-market effects.

6.4 Main limitations

The model abstracts from nontradable and intermediate goods, plant-level fixed costs in the baseline, arm's-length offshoring outside multinational firms, and rich dynamics of innovation accumulation. These abstractions affect the magnitude of gains, but the paper argues that the qualitative cross-country logic is robust.

Interpretation: The empirical design is credible because different moments discipline different mechanisms. Trade shares discipline goods-market frictions, MP shares discipline technology-use frictions, and restricted gravity disciplines productivity correlation.

7 Section-by-section study guide

- **Introduction.** Establishes the empirical pattern that innovative countries tend to have outward MP and motivates fears about production relocation.
- **Model.** Defines CES demand, firm innovation, productivity draws, MP and trade costs, and the location decisions of firms.
- **Comparative statics and welfare.** Shows how trade and MP affect specialization, innovation shares, and welfare through direct and indirect channels.
- **Quantification.** Uses trade, MP, production, and gravity moments to calibrate the model.
- **Fit and implications.** Compares model-generated trade, MP, and innovation shares to observed data.
- **Counterfactuals.** Studies gains from openness, MP liberalization, China integration, and robustness checks.
- **Conclusion.** Emphasizes that openness reorganizes innovation and production but does not generally support the fear that production workers in innovating countries lose.

8 Tables and figures

8.1 Figure 1: R&D and Net Multinational Production + Figure 2: Manufacturing R&D and Employment of US Multinational Firms

Read:

- Figure 1 plots manufacturing R&D intensity against net MP. Countries with higher R&D/value added tend to have higher outward relative to inward MP.
- The United States, Japan, Finland, Sweden, Germany, and Benelux lie toward the innovation/outward-MP side of the diagram.
- Countries with lower R&D intensity tend to be less outward-MP oriented or more production-host oriented.
- Figure 2 shows the US time-series fact: from 1999 to 2009, manufacturing R&D intensity rose and the foreign-affiliate employment share of US firms also rose.
- The Chinese affiliate employment share in total US foreign affiliate employment rose sharply, emphasizing China's role in the reallocation of production.

Detailed interpretation: These figures motivate the entire model. Figure 1 is cross-sectional: innovative countries are also net exporters of multinational production services. Figure 2 is time-series: the United States becomes more innovation-intensive while also moving more employment to foreign affiliates. Together, the two figures show why a model must allow innovation and production to separate geographically.

Economic message: The data suggest that ideas and production can be geographically separated. Multinational production is the mechanism through which innovative countries can export ideas while importing physical production.

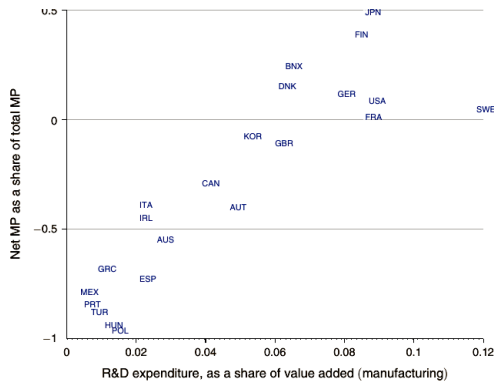


FIGURE 1. R&D AND NET MULTINATIONAL PRODUCTION

(a) Figure 1: R&D and Net Multinational Production

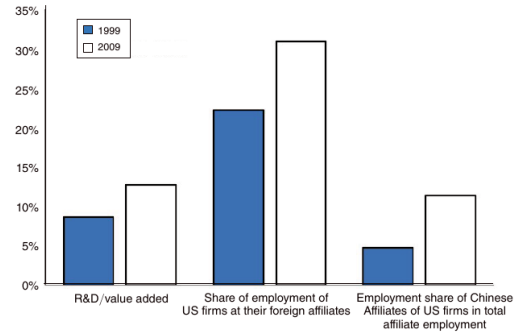


FIGURE 2. MANUFACTURING R&D AND EMPLOYMENT OF US MULTINATIONAL FIRMS

(b) Figure 2: Manufacturing R&D and Employment of US Multinational Firms

8.2 Figure 3: Firm Costs and Pricing Behavior + Figure 4: Multivariate Pareto: Simulated Draws

Read:

- Figure 3 summarizes the core cost chain: innovation in country i , production in country l , and consumption in country n .
- The MP cost γ_{il} links innovation origin to production location; the trade cost τ_{ln} links production location to consumption market.
- Figure 4 shows how the productivity correlation parameter ρ changes the joint distribution of productivity draws across production locations.
- With low ρ , draws are weakly correlated and production advantages are more location-specific.
- With high ρ , high productivity in one location tends to be associated with high productivity in another location.

Detailed interpretation: Figure 3 is the model's map: idea origin, factory location, and market destination are distinct. Figure 4 explains why restricted gravity helps identify ρ : if firms' productivity draws are correlated across locations, then trade flows within multinational networks respond differently to trade costs than aggregate trade flows.

Economic message: The model's novelty comes from letting firms choose production locations after innovation. The multivariate productivity structure makes that choice tractable in a many-country model.

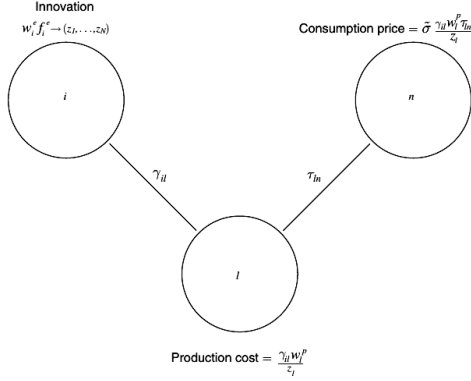


FIGURE 3. FIRM COSTS AND PRICING BEHAVIOR

Notes: Innovation is done in country i at cost $w_{il}f_i^\kappa$; production is done in country l at unit cost $\gamma_l w_l^\rho / z_l$; and consumption is done in country n at price $\tilde{\sigma} \gamma_n w_n^\rho / z_n$.

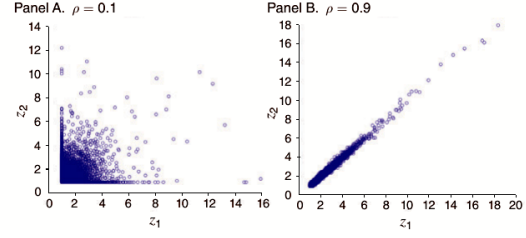


FIGURE 4. MULTIVARIATE PARETO: SIMULATED DRAWS

∴ Simulation for 100,000 draws, $N = 2$, $\theta = 4$, and $T_1 = T_2$.

(a) Figure 3: Firm Costs and Pricing Behavior

(b) Figure 4: Multivariate Pareto: Simulated Draws

8.3 Table 1: Restricted and Unrestricted Gravity + Table 2: Calibrated Model Parameters and Data Targets

Read:

- Table 1 shows that restricted trade flows are more sensitive to tariffs than unrestricted flows.
- In the OLS unweighted specification, the restricted tariff coefficient is about -10.9 while the unrestricted coefficient is about -4.3 .
- In the weighted and PPML specifications, the restricted elasticity remains substantially larger in absolute value.
- Table 2 summarizes the calibration. The paper sets $\sigma = 4$, $\kappa = 2$, $\theta/(1 - \rho) = 10$, $\theta = 4.5$, $\rho = 0.55$, and $\eta = 0.167$.
- The calibration targets markups, trade elasticities, profit shares, production and innovation productivity, trade shares, and MP shares.

Detailed interpretation: Table 1 is not just a preliminary regression. It is a key source of identification for the multivariate productivity distribution. Table 2 then turns these moments into model parameters. Restricted gravity pins down the combination $\theta/(1 - \rho)$, while unrestricted gravity helps pin down θ and therefore ρ .

Economic message: The model is disciplined by a moment that is specific to multinational networks, not just aggregate trade. This gives the quantitative exercise leverage on MP-specific mechanisms.

Tariffs:	OLS		PPML	
	Unweighted average	Weighted average	Unweighted average	Weighted average
Restricted	−10.9 (3.5)	−11.1 (3.5)	−8.4 (2.6)	−11.6 (4.0)
Unrestricted	−4.3 (1.8)	−5.3 (1.9)	−5.4 (1.7)	−8.6 (2.6)

Notes: Unweighted average refers to a simple average across industry tariffs; weighted average refers to an average across industry tariffs using as weights the share of the industry in total trade. The number of observations is 317 in the OLS specification and 384 in the PPML specification. Robust standard errors in parentheses.

(a) Table 1: Restricted and Unrestricted Gravity

Notation	Value	Parameters	Moments
		Description	Description
σ	4	Elasticity of substitution	Mark-up (OECD)
κ	2	Fréchet shape parameter	Literature
$\theta/(1 - \rho)$	10	Restricted trade elasticity	Restricted gravity equation
θ	4.5	MVP shape parameter	Unrestricted gravity equation (US)
ρ	0.55	MVP correlation parameter	Implied from restr. gravity and θ
η	0.167	Profit share	Implied from θ and σ
T_i^p	0.38 (0.44)	Average productivity in production in i	Gross manufacturing production in i
T_i^e	1.77 (1.13)	Average productivity in innovation in i	Innovation rate in i
$\hat{\tau}_{ln}^p$	2.9 (0.8)	Trade cost from l to n	Trade share from l to n
$\hat{\tau}_{il}^p$	4.4 (3.5)	MP cost from i to l	MP share from i to l

Notes: Trade elasticity refers to the elasticity of exports of firms from i located in l and selling to n to trade costs from l to n . Parameter values for T_i^p and T_i^e refer to averages across N countries, relative to values for the United States. Parameter values for τ_{ln} and τ_{il} refer to averages across the $N \times (N - 1)$ country pairs. Standard deviations are in parentheses.

(b) Table 2: Calibrated Model Parameters and Data Targets

8.4 Table 3: Comparative Advantage versus Home-Market Effects + Table 4: Trade Costs, MP Costs, and Gravity

Read:

- Table 3 reports production productivity T_i^p , innovation productivity T_i^e , and innovation shares r_i for each country.
- The baseline innovation share separates innovation-specializing countries from production-specializing countries relative to $\eta = 0.167$.
- Column 4 shuts down comparative advantage by setting $T_i^e = A_i = 1$, isolating the home-market effect component of innovation specialization.
- Table 4 shows that estimated trade and MP costs rise with distance and tend to be lower for countries sharing a language.
- The R^2 is especially high for bilateral MP costs, indicating that geographic variables explain much of the estimated MP cost variation.

Detailed interpretation: Table 3 is the bridge between data and interpretation. It separates countries that innovate because they are intrinsically productive in innovation from countries that innovate because home-market effects make innovation attractive. Table 4 then checks whether calibrated trade and MP costs behave like standard gravity frictions.

Economic message: The paper’s specialization mechanism combines comparative advantage with market size. Countries can specialize in innovation either because they are good at innovation or because their market position makes innovation profitable.

TABLE 3—COMPARATIVE ADVANTAGE VERSUS HOME-MARKET EFFECTS

	T_i^p (1)	T_i^s (2)	Innovation share, r_i	
			Baseline (3)	$T_i^s = A_i = 1$ (4)
Australia (AUS)	0.175	0.695	0.067	0.086
Austria (AUT)	0.226	2.626	0.147	0.073
Benelux (BNX)	0.600	1.549	0.203	0.183
Brazil (BRA)	0.206	0.745	0.145	0.154
Canada (CAN)	0.403	1.371	0.129	0.097
China (CHN)	0.003	0.140	0.139	0.167
Cyprus (CYP)	0.036	3.826	0.167	0.167
Denmark (DNK)	0.162	3.573	0.193	0.097
Spain (ESP)	0.568	1.660	0.137	0.125
Finland (FIN)	0.247	4.136	0.195	0.084
France (FRA)	0.589	1.740	0.175	0.138
United Kingdom (GBR)	0.523	1.465	0.159	0.131
Germany (GER)	0.480	1.132	0.174	0.184
Greece (GRC)	0.178	3.832	0.160	0.131
Hungary (HUN)	0.051	1.247	0.059	0.050
Ireland (IRL)	0.240	2.104	0.078	0.033
Italy (ITA)	1.176	2.007	0.151	0.130
Japan (JPN)	1.963	1.931	0.178	0.159
Korea (KOR)	0.242	1.689	0.162	0.144
Mexico (MEX)	0.006	0.460	0.133	0.147
Poland (POL)	0.001	0.371	0.121	0.151
Portugal (PRT)	0.142	1.565	0.097	0.086
Romania (ROM)	0.0001	0.301	0.159	0.165
Sweden (SWE)	0.332	2.708	0.162	0.091
Turkey (TUR)	0.091	1.163	0.159	0.159
United States (USA)	1.000	1.000	0.182	0.202

Notes: Columns 1 and 2 shows the values for T_i^p and T_i^s , respectively, from the baseline calibration. Columns 3 and 4 show innovation shares, r_i , coming from the model's equilibria with T_i^s and T_i^p set to their baseline values, and

(a) Table 3: Comparative Advantage versus Home-Market Effects

TABLE 4—TRADE COSTS, MP COSTS, AND GRAVITY: BASELINE CALIBRATION

	Distance dummies					Other gravity controls		
	D2	D3	D4	D5	D6	Border	Language	R^2
Bilateral trade costs	0.966 (0.10)	0.954 (0.10)	1.112 (0.08)	1.159 (0.09)	1.149 (0.06)	0.035 (0.04)	-0.091 (0.05)	0.61
Bilateral MP costs	0.702 (0.09)	0.882 (0.1)	1.049 (0.09)	1.091 (0.09)	1.235 (0.06)	-0.065 (0.04)	-0.154 (0.03)	0.93

Notes: Coefficients are estimated through OLS. Robust standard errors in parentheses.

(b) Table 4: Trade Costs, MP Costs, and Gravity

8.5 Figure 5: Aggregate Trade and MP Shares + Figure 6: Bilateral Trade and MP Shares

Read:

- Figure 5 compares aggregate model-implied outward MP, inward MP, exports, and imports with observed data.
- The model captures the broad cross-country ranking of aggregate trade and MP shares.
- Figure 6 compares bilateral trade and MP shares in logs.
- Bilateral MP shares fit very tightly, while bilateral trade shares are noisier but still positively related to the data.
- The figures show that the calibration is overidentified: it cannot mechanically fit every bilateral object perfectly.

Detailed interpretation: Figure 5 checks country-level fit; Figure 6 checks bilateral fit. The strong bilateral MP fit is important because MP is the channel that distinguishes this model from standard trade frameworks. The less perfect trade fit reflects the parsimonious symmetric-cost assumptions.

Economic message: The model is not just matching total openness. It captures enough bilateral structure to make counterfactual specialization exercises credible.

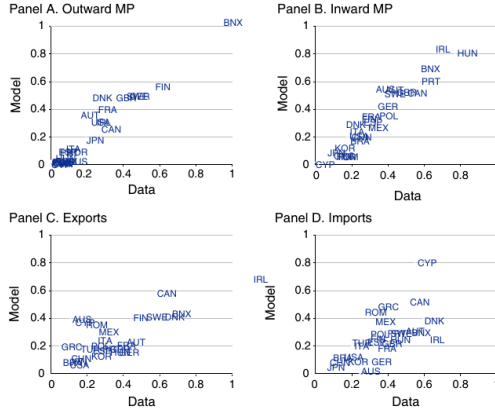


FIGURE 5. AGGREGATE TRADE AND MP SHARES: MODEL VERSUS DATA

Notes: Imports and exports for country n are normalized by manufacturing absorption in country n . Outward (inward) MP shares refer to total sales of foreign affiliates from (into) country n normalized by gross production in country n in the model (GDP in country n in the data).

(a) Figure 5: Aggregate Trade and MP Shares: Model versus Data

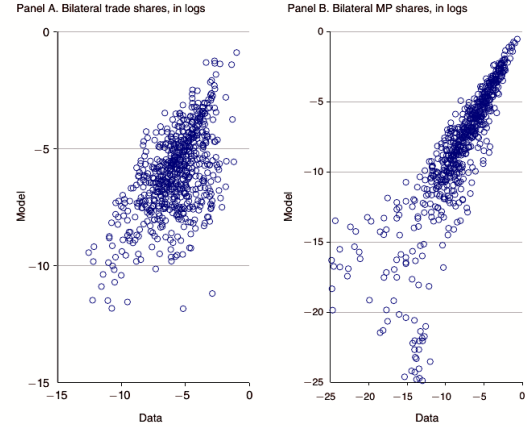


FIGURE 6. BILATERAL TRADE AND MP SHARES: MODEL VERSUS DATA

otes: Imports and exports from country l to n are normalized by manufacturing absorption in country n . MP flow

(b) Figure 6: Bilateral Trade and MP Shares: Model versus Data

8.6 Figure 7: Innovation Shares + Table 5: Gains from Openness, Trade, and MP

Read:

- Figure 7 compares model-implied innovation shares with R&D employment shares in the data, both relative to the United States.
- The model and data are positively correlated, even though measured R&D employment is a narrower concept than model innovation.
- Table 5 decomposes gains from openness into direct and indirect components, and separately reports gains from trade and gains from MP.
- All selected countries gain from openness, but some have negative indirect effects.
- Some countries have negative gains from trade or MP alone, highlighting that partial liberalizations can reallocate innovation in welfare-relevant ways.

Detailed interpretation: Figure 7 validates the innovation side of the model with data not directly targeted in the same way as trade and MP flows. Table 5 shows the welfare logic: direct access gains dominate, but indirect innovation-reallocation effects can be negative. Ireland is a useful example: direct gains are very high, but indirect losses offset part of them.

Economic message: Openness raises welfare overall, but gains are not just price-index effects. When innovation moves, terms of trade and worker income distribution also move.

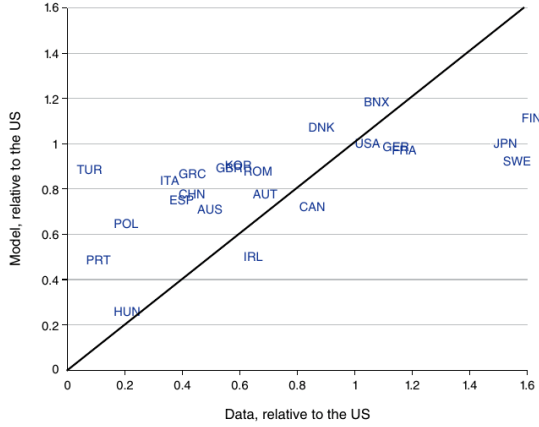


FIGURE 7. INNOVATION SHARES: MODEL VERSUS DATA

s: Innovation shares refer to R&D employment shares: in the model, the (equilibrium) variable r , calculated exogenously current account imbalances, $\Delta \neq 0$ (the variable in column 3 of Table 3); in the data, manufacturing R&D employment as a share of total manufacturing employment, closest year available to 1999, from OECD data on total manufacturing employment for China are from Lardy 2015). Brazil, Mexico, and Cyprus do not have data.

(a) Figure 7: Innovation Shares: Model versus Data

TABLE 5—GAINS FROM OPENNESS, TRADE, AND MP: SELECTED COUNTRIES, BASELINE CALIBRATION (Percent Change)

	Gains from openness			Gains from trade (4)	Gains from MP (5)
	Overall (1)	Direct (2)	Indirect (3)		
Australia	20.7	42.2	-15.2	-7.6	11.7
Benelux	60.2	52.9	4.8	9.7	27.8
China	3.4	7.4	-3.7	-2.2	0.04
Germany	18.2	17.1	0.9	2.5	10.2
Hungary	44.0	73.0	-16.7	-5.2	16.3
Ireland	89.5	118.2	-13.2	-1.0	27.7
Mexico	16.7	22.4	-4.7	1.8	1.0
Turkey	5.8	6.8	-1.0	2.6	-0.5
United States	9.8	7.6	2.0	3.2	-5.3
Average (all sample)	24.6	29.0	-2.9	3.6	7.0

Note: The gains from openness refer to changes in real expenditure between autarky and the calibrated equilibrium. The direct and indirect effects refer to the first and second terms, respectively, on the right-hand side of (27). The gains from trade (MP) refer to changes in real expenditure between an equilibrium with only MP (trade) and the calibrated equilibrium.

(b) Table 5: Gains from Openness, Trade, and MP

8.7 Table 6: MP Liberalization + Table 7: The Rise of the East

Read:

- Table 6 studies a 5 percent decrease in all MP costs. The average country experiences a rise in real expenditure, but innovation shares can fall.
- Production wages tend to rise more broadly than innovation wages in some countries, highlighting distributional effects across worker types.
- Table 7 studies China-related counterfactuals: China in autarky, unilateral MP liberalization into China, and frictionless US MP into China.
- China in autarky raises China's innovation share but lowers real expenditure; global welfare generally falls when China is removed from openness.
- Frictionless US MP into China dramatically shifts innovation activity toward the United States and away from China.

Detailed interpretation: Table 6 shows that lowering MP barriers does not uniformly raise innovation everywhere. It can push production-specializing countries further into production. Table 7 shows that China's integration matters through both production opportunities and innovation reallocation. The largest effects occur when US firms can produce in China without MP frictions.

Economic message: MP liberalization is not simply “good for capital and bad for labor.” Its effects depend on whether a country becomes an innovation hub, a production location, or a source of ideas used abroad.

TABLE 6—MP LIBERALIZATION: SELECTED COUNTRIES,
BASELINE CALIBRATION (Percent Change)

	r (1)	X/P (2)	w^p/P (3)	w^e/P (4)
Australia	-11.49	3.40	3.83	-2.72
Benelux	10.63	5.61	4.17	11.08
China	-5.33	0.26	0.70	-2.45
Germany	1.29	2.71	2.57	3.37
Hungary	-14.80	3.76	4.24	-4.23
Ireland	7.52	4.94	4.60	8.81
Mexico	-8.26	0.81	1.45	-3.44
Turkey	-2.14	-0.01	0.20	-1.08
United States	0.75	1.32	1.23	1.70
Average (all sample)	-2.00	1.93	2.01	0.88

Notes: MP liberalization refers to a 5 percent decrease in all MP costs with respect to the baseline calibrated values. The variables are: the innovation share, r ; real expenditure, X/P ; real wage per efficiency unit in the production sector, w^p/P ; real wage per efficiency unit in the innovation sector, w^e/P . Percentage changes are with respect to the baseline calibrated equilibrium without trade and MP imbalances, $\Delta = 0$.

(a) Table 6: MP Liberalization

TABLE 7—THE RISE OF THE EAST: BASELINE CALIBRATION

	Percent changes				
	China	United States	Japan	Mexico	Ireland
<i>China in autarky</i>					
Innovation rate, r	19.55	-2.53	-1.81	0.34	-0.79
Real expenditure, X/P	-3.29	-0.76	-0.76	-0.11	-0.37
Real production wage, w^p/P	-4.84	-0.48	-0.57	-0.14	-0.33
Real innovation wage, w^e/P	5.74	-2.03	-1.67	0.06	-0.76
<i>Unilateral MP liberalization into China</i>					
Innovation rate, r	-11.95	1.34	1.21	-0.25	0.58
Real expenditure, X/P	0.75	0.31	0.26	0.013	0.14
Real production wage, w^p/P	1.73	0.16	0.13	0.03	0.12
Real innovation wage, w^e/P	-5.45	0.98	0.86	-0.11	0.43
<i>Frictionless MP into China from USA</i>					
Innovation rate, r	-92.25	21.87	-0.86	-3.36	-5.56
Real expenditure, X/P	34.20	4.85	0.0004	0.33	0.68
Real production wage, w^p/P	43.88	2.27	0.09	0.59	0.92
Real innovation wage, w^e/P	-62.65	15.75	-0.43	-1.36	-2.16

Notes: China in autarky refers to the counterfactual scenario in which trade and MP costs from/to China are set to infinity; unilateral MP liberalization into China refers to the counterfactual scenario in which MP costs into China are decreased by 10 percent; frictionless MP into China from USA refers to the counterfactual scenario in which MP costs from the United States into China are set to zero. The real wage in each sector refers to the real wage

(b) Table 7: The Rise of the East

9 Detailed result map

- **Stylized fact.** Innovative countries are net exporters of multinational production activity.
- **Model structure.** Firms choose production locations separately from innovation locations, creating a three-country structure (i, l, n) .
- **Identification result.** Restricted trade elasticities are larger than unrestricted elasticities, supporting a correlated multivariate productivity structure.
- **Calibration result.** The model matches aggregate and bilateral trade/MP shares and produces innovation shares positively associated with R&D employment shares.
- **Openness result.** Gains from openness are positive for all countries in the selected set, but indirect innovation-reallocation effects can reduce the net gains.
- **Trade/MP substitution.** Gains from trade and gains from MP alone are smaller than gains from total openness because trade and MP are substitutes.
- **MP liberalization result.** Lower MP costs raise average real expenditure but shift specialization patterns and wages across innovation and production sectors.
- **China result.** China's integration reallocates innovation and production globally; countries linked to China can benefit disproportionately.

10 Short summary

- The paper studies globalization as the joint opening of goods trade and multinational production.
- It distinguishes idea creation from physical production.
- The model embeds endogenous innovation in a many-country trade and MP environment.
- Countries can specialize in innovation because of comparative advantage or home-market effects.
- Multinational production lets innovative countries export ideas and import goods.
- The calibration uses trade, MP, production, R&D, and gravity data for 26 countries.
- A key empirical moment is that restricted multinational-network trade flows are more elastic to trade costs than unrestricted aggregate flows.
- The model fits aggregate and bilateral MP shares reasonably well.

- All countries gain from overall openness, but partial trade or MP liberalization can create small losses for some countries.
- MP liberalization shifts innovation and production shares across countries.
- China integration reallocates innovation and production globally.
- Production workers need not lose from offshoring because price effects and general-equilibrium wage effects can dominate production relocation.

11 Conclusion

11.1 Core conclusion

Arkolakis, Ramondo, Rodríguez-Clare, and Yeaple show that globalization changes more than the cost of buying goods. By allowing production to move away from the country where ideas are created, multinational production changes the global allocation of innovation, production, profits, and wages. Their model turns this mechanism into a quantifiable framework that can be calibrated to real trade and MP flows.

11.2 Economic intuition

The intuition is that firms choose both where to create ideas and where to use them. If MP costs are high, ideas must be used close to home, so production and innovation are geographically tied. If MP costs fall, firms in innovation-intensive countries can move production abroad while retaining the profits from ideas. This can strengthen innovation specialization in some countries and production specialization in others.

11.3 What is new relative to standard trade models

A standard trade model can explain gains from access to cheaper goods, but it cannot explain why innovation moves across countries when MP becomes easier. This paper adds the missing link: the location of production affects the return to creating firms. Therefore, trade and MP policies alter innovation incentives and not only goods prices.

11.4 Policy implication

The paper cautions against treating offshoring or MP liberalization as a simple zero-sum conflict between countries. The gains are generally positive and broad, but the distribution across countries and worker types depends on whether liberalization shifts a country toward innovation or production. Policies aimed at innovation capacity, worker mobility, and intangible investment matter for how countries benefit from MP.

11.5 Limitations

The quantitative model abstracts from nontradable goods, intermediate input chains, plant-level fixed costs in the baseline, arm's-length offshoring, and dynamic accumulation of knowledge. The authors discuss some robustness, but these features could matter for the magnitude of welfare effects.

11.6 Takeaway

The paper's main lesson is that globalization should be analyzed as a joint system of trade in goods and movement of production technology. Countries do not simply export and import goods; they also export

and import the use of ideas. This distinction is essential for understanding why globalization can increase specialization in innovation and production while still generating aggregate gains.